QR Code-Enabled Web-Based Resource Borrowing and Returning System with Availability Tracking and Faculty Authentication for MIS Faculty at ICCT Cainta Campus

(Revised to Reflect Deployed System — May 2026)

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Revised Documentation

Based on the Deployed System at **https://icct-mis.classapparelph.com**

Date of Revision: May 23, 2026

ABSTRACT

The borrowing and returning of resources at ICCT Cainta Campus currently relies on manual request letters or faculty ID presentation, which is inefficient, time-consuming, and lacks transparency in tracking availability. Professors often experience delays in borrowing, while MIS faculty struggle with monitoring overdue returns and managing resource quantities. This study developed and deployed a web-based automated resource borrowing and returning system that replaces the manual request process and streamlines transactions. Faculty members access the system through a web-based dashboard via any browser using email and password authentication, where they can view resource availability and submit digital borrowing requests. While originally designed with QR code integration as a core feature, the development process prioritized stable and accessible web-based authentication to ensure immediate usability across all devices without requiring QR scanning hardware. Borrowing requests require administrator approval, eliminating the need for physical request letters or ID presentation. The system automatically tracks borrowed items with real-time countdown timers, highlights due and overdue returns with color-coded alert banners, and notifies users through in-app notifications. Additional features implemented during development include a room/location tracking field for borrowed resources, a settings system for runtime configuration, automated due/overdue monitoring via server-side cron job with intelligent deduplication, and a REST API for future mobile application development. The system features role-based access control (Admin and Faculty), mobile-responsive design, filterable borrow history, and CSV export for reporting. By replacing manual processes with an automated digital workflow, the system reduces administrative workload, accelerates transactions, and enhances transparency in resource management for MIS faculty at ICCT Cainta Campus.

Keywords* — *QR Code Technology, Web-Based System, Resource Management, Automated Borrowing and Returning, REST API, Information System

CHAPTER I: INTRODUCTION

A. Background and Motivation

Resource borrowing and returning is an essential support function in academic institutions, particularly for faculty members who rely on shared materials and equipment to deliver instruction effectively. At ICCT Cainta Campus, professors in the MIS department currently request resources either by submitting a formal request letter or by presenting their faculty ID. While this process ensures accountability, it is slow, manual, and prone to delays. MIS faculty also face challenges in monitoring overdue returns and tracking the availability of resources, as there is no centralized system that provides real-time visibility of inventory. These inefficiencies disrupt schedules, reduce productivity, and complicate resource management. In the field of Information Technology, automation and digital systems have been widely adopted to streamline workflows, improve accuracy, and enhance transparency. This motivated the development of a QR code-enabled web-based system that modernizes the borrowing and returning process for MIS faculty at ICCT Cainta Campus. During development, the system evolved to prioritize web-based authentication and a broader feature set while maintaining the original design goals of eliminating manual processes and improving transparency.

B. Problem Statement

The existing borrowing and returning process at ICCT Cainta Campus is inefficient and outdated. Professors must either submit a request letter or present their faculty ID to borrow resources, which consumes time and lacks transparency. MIS faculty struggle to monitor overdue returns and resource availability, as there is no centralized dashboard to track inventory in real time. These limitations result in delays, poor accountability, and reduced efficiency in resource management.

C. Objectives of the Study

The general objective of this study is to design and develop a web-based system that automates the borrowing and returning of resources, replacing the current manual process of request letters and faculty ID presentation. Specifically, the study achieved the following objectives:

1. Provide a web-based dashboard accessible via browser URL where professors can log in using email and password authentication, with the option for QR code integration for streamlined access in future iterations.
2. Display available resources and their quantities in real time with visual availability indicators.
3. Allow professors to submit borrowing requests digitally, eliminating the need for request letters or physical ID presentation, with duration-based return times and room/location tracking.
4. Enable administrators to approve or reject borrowing requests and monitor due and overdue returns through a color-coded dashboard with countdown timers.
5. Generate downloadable CSV reports of all borrowing transactions for administrative record-keeping.
6. Implement a room designation field to track the physical location of borrowed resources.
7. Provide in-app notifications for approvals, rejections, due dates, and overdue items with unread badge tracking.
8. Develop automated due and overdue monitoring through server-side cron jobs with intelligent deduplication to prevent notification spam.
9. Design a complete RESTful API (Sanctum) for future mobile application (Flutter) development, along with a settings system for runtime configuration.

D. Scope of the Study

The scope defines the specific areas, users, and functionalities covered by the system:

- **Target Users:** The system is designed for the MIS faculty and administrators at the ICCT Cainta Campus.
- **Authentication:** Users access the system via email and password login with role-based access control (Admin and Faculty). Password recovery is handled through Brevo transactional email service, eliminating the need for manual administrator intervention.
- **Real-Time Resource Management:** The system provides real-time tracking and display of available resources and their exact quantities with visual availability indicators.

- **Digital Workflow:** Professors can submit borrowing requests digitally with duration-based return times (1 hour, 2 hours, 3 hours, or custom) and a room/location field.
- **Administrative Oversight:** Administrators approve or reject requests, mark items as returned, and monitor due/overdue items through color-coded countdown timers and alert banners.
- **Automated Notifications:** In-app notifications with unread badge tracking notify users and administrators regarding approvals, rejections, due dates, and overdue items.
- **Automated Monitoring:** A server-side cron job (borrows:check-due) runs every minute, performing dedicated checks for items due now, due within 30 minutes, and overdue items, with intelligent deduplication.
- **Borrow History:** Complete borrow history with filterable tabs (All, Pending, Approved, Borrowed, Returned, Rejected), summary statistics, and color-coded status indicators.
- **CSV Export:** Administrators can download complete borrowing transaction records in CSV format.
- **REST API:** Complete RESTful API secured with Laravel Sanctum token authentication, supporting authentication, dashboard, resources, borrows, notifications, profile, and user management endpoints.
- **Settings System:** Key-value database configuration store for runtime settings such as auto-dismiss notification timing.
- **QR Code Integration:** The original design includes QR code scanning as a foundational concept. Web-based authentication was prioritized during development to ensure immediate device compatibility, with QR capabilities available for future enhancement.
- **Technical Platform:** Laravel 11 with MySQL database, Apache 2.4 web server with HTTPS (Let's Encrypt SSL), Tailwind CSS with Alpine.js for frontend, Vite for asset bundling, and Brevo (formerly SendinBlue) for transactional email service. Hosted on Ubuntu 22.04 LTS with adequate server resources.

E. Limitations

- **Connectivity Requirements:** The system requires internet connectivity to function.
- **Account Credential Dependency:** Password recovery requires access to the registered email account.

- **Hardware Dependency:** Users need a device with a modern web browser.
- **Maintenance Boundaries:** The system excludes physical maintenance of resources; it only tracks digital status and availability.
- **Infrastructure Constraints:** May face challenges with bandwidth limitations or older browsers.
- **Geographic and Departmental Focus:** Deployed for ICCT Cainta MIS department only; not yet scaled to other departments or locations.
- **Data Integrity:** Relies on borrower honesty to report resource condition upon return.
- **Notification Scope:** All alerts are delivered as in-app notifications. Users must be logged in to receive them.
- **Mobile App:** A native mobile application (Flutter) is not yet developed; currently web-only.

CHAPTER II: RELATED WORKS

The continuous advancement of information technology has reshaped how academic institutions manage student identification, attendance monitoring, and resource utilization. Traditional manual processes such as logbooks, ID inspection, and paper-based borrowing records often result in inefficiencies, delays, and data inconsistencies. Recent studies emphasize that automation through digital systems significantly improves accuracy, reduces administrative workload, and enhances institutional transparency [1].

QR code technology has become one of the most practical tools for implementing automated monitoring systems in educational environments. According to Roy [1], QR-based attendance systems offer a low-cost yet reliable alternative to manual attendance tracking. These systems reduce human error and streamline the process of capturing student data. Similarly, Masalha and Hirzallah [2] demonstrated that QR code scanning enables faster attendance recording while maintaining data accuracy and accessibility for administrators.

In addition to attendance monitoring, QR codes have been applied in broader academic resource management contexts. Rahman et al. [3] explain that QR technology can be effectively used for tracking borrowed materials such as laboratory tools and library books. Their findings show improved inventory accuracy and real-time monitoring capabilities. By digitizing borrowing records, institutions gain structured data that can support planning and decision-making processes.

The effectiveness of QR systems also depends on system design and quality standards. Liew and Tan [4] highlighted the importance of authentication accuracy and fraud prevention in QR-based attendance applications. Meanwhile, Sharma et al. [5] stressed that academic information systems must comply with quality models such as ISO/IEC 25010 to ensure usability, reliability, performance efficiency, and compatibility across devices. Without proper system design considerations, digital platforms may face adoption challenges among users.

Several foreign studies have explored integrated systems that combine login authentication with resource management functions. Singh et al. [11] developed an access control and resource tracking system that reduced manual processing time and minimized record errors. Hariyanto et al. [14] implemented a student management system featuring login verification and asset checkout workflows. Mahmood et al. [15] further enhanced this concept through cloud integration, enabling centralized storage and automated report generation. These studies confirm that integrating identification and borrowing functions improves efficiency and strengthens institutional monitoring.

Within the Philippine context, digital and automated systems have been widely adopted for attendance monitoring and ID validation. Local studies report improvements in efficiency, user acceptability, and administrative workload reduction [6], [7]. Research on digital attendance monitoring also highlights enhanced punctuality and accountability among users [8], [10]. Additionally, digital ID validation systems have demonstrated faster verification procedures and improved security compared to traditional manual inspection methods [9]. An integrated library circulation system implemented in a local campus further showed improved transaction speed and record accuracy [20].

Modern information systems increasingly adopt API-first architectures to support multi-platform access [21]. By exposing system functionality through RESTful APIs, institutions can develop mobile applications and third-party integrations that extend the reach of the core system while maintaining consistent business logic and security. Automated scheduling through cron jobs further enhances system reliability by enabling routine maintenance tasks such as due-date monitoring without manual intervention [22].

Research Gap

Existing studies have demonstrated the effectiveness of automated systems for attendance monitoring, library circulation, and resource tracking. However, most implementations focus on isolated functions without providing a comprehensive solution that integrates faculty authentication, resource availability tracking, digital borrowing requests with duration-based return times and room designation, administrative approval workflows, automated overdue monitoring via cron scheduling with notification deduplication, in-app notification systems, CSV reporting, and REST API support on a single platform. This gap highlights the need for a centralized web-based faculty resource borrowing management system with comprehensive automation capabilities, as implemented for ICCT Cainta Campus.

CHAPTER III: METHODOLOGY

A. Research Design

This study adopted a developmental research design focusing on systematic analysis, design, and implementation of an information system. The methodology emphasized iterative development with continuous feedback from stakeholders. Following the SDLC framework, the system progressed through planning, analysis, design, development, testing, and deployment phases, resulting in a fully functional live system accessible at https://icct-mis.classapparelph.com.

B. System Architecture

The system follows a three-tier architecture supporting both web and mobile clients, with a Laravel 11 backend, MySQL database, Brevo email service, and cron-based task scheduling, all deployed on an Apache 2.4 web server with HTTPS security.

Fig. 1 — System Architecture Diagram of the Deployed System

C. Conceptual Framework

The proposed system follows an enhanced Input–Process–Output (IPO) model with a feedback loop to support continuous improvement and monitoring. The Input section includes faculty credentials, resource inventory data, device access, and administrator control parameters. The Process section encompasses authentication, digital borrowing request submission, room designation, duration calculation, administrator approval workflows, transaction recording, real-time inventory updates, cron-based due-date monitoring, notification generation, and report compilation. The Output section presents authenticated transactions, approved borrowings, updated resource availability, overdue alerts, and generated usage reports. The Feedback component provides borrowing trends, resource utilization insights, and system performance data for continuous improvement.

D. Database Schema

The database consists of 6 tables: users, resources, borrows, notifications, settings, and personal_access_tokens (Sanctum). The borrows table includes duration (in minutes), room/location, and status tracking fields.

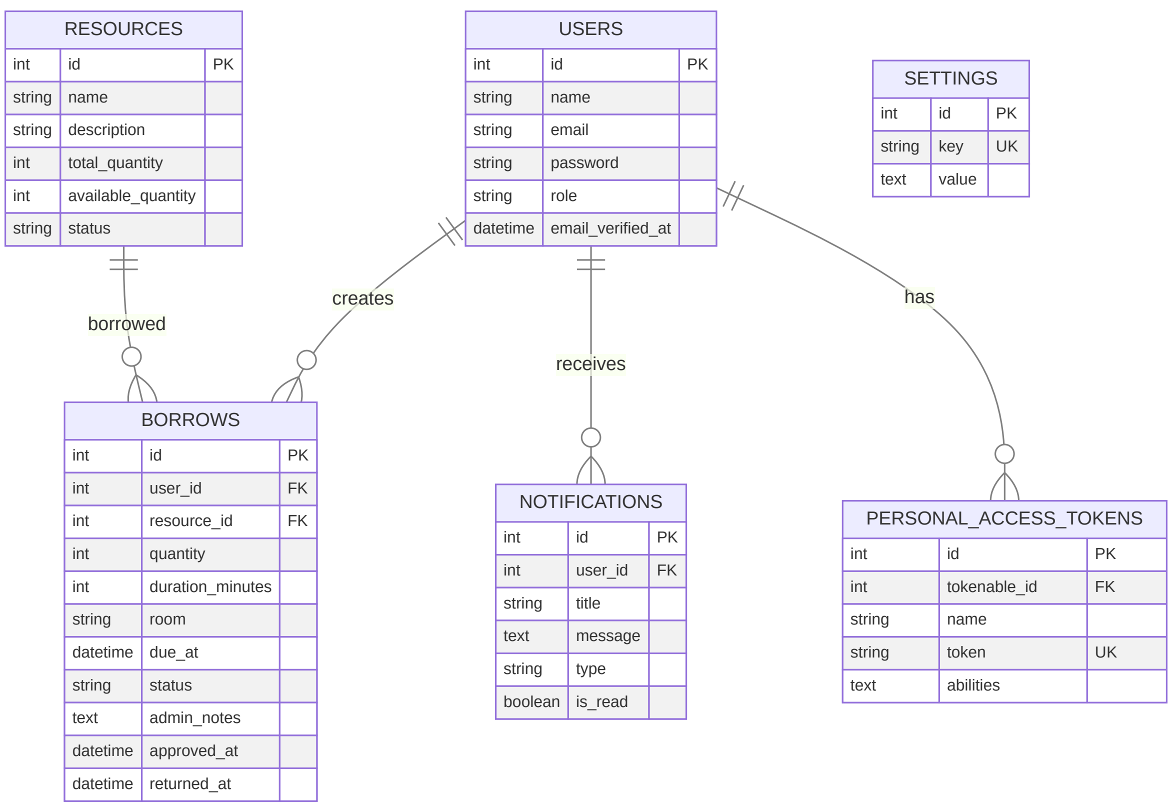


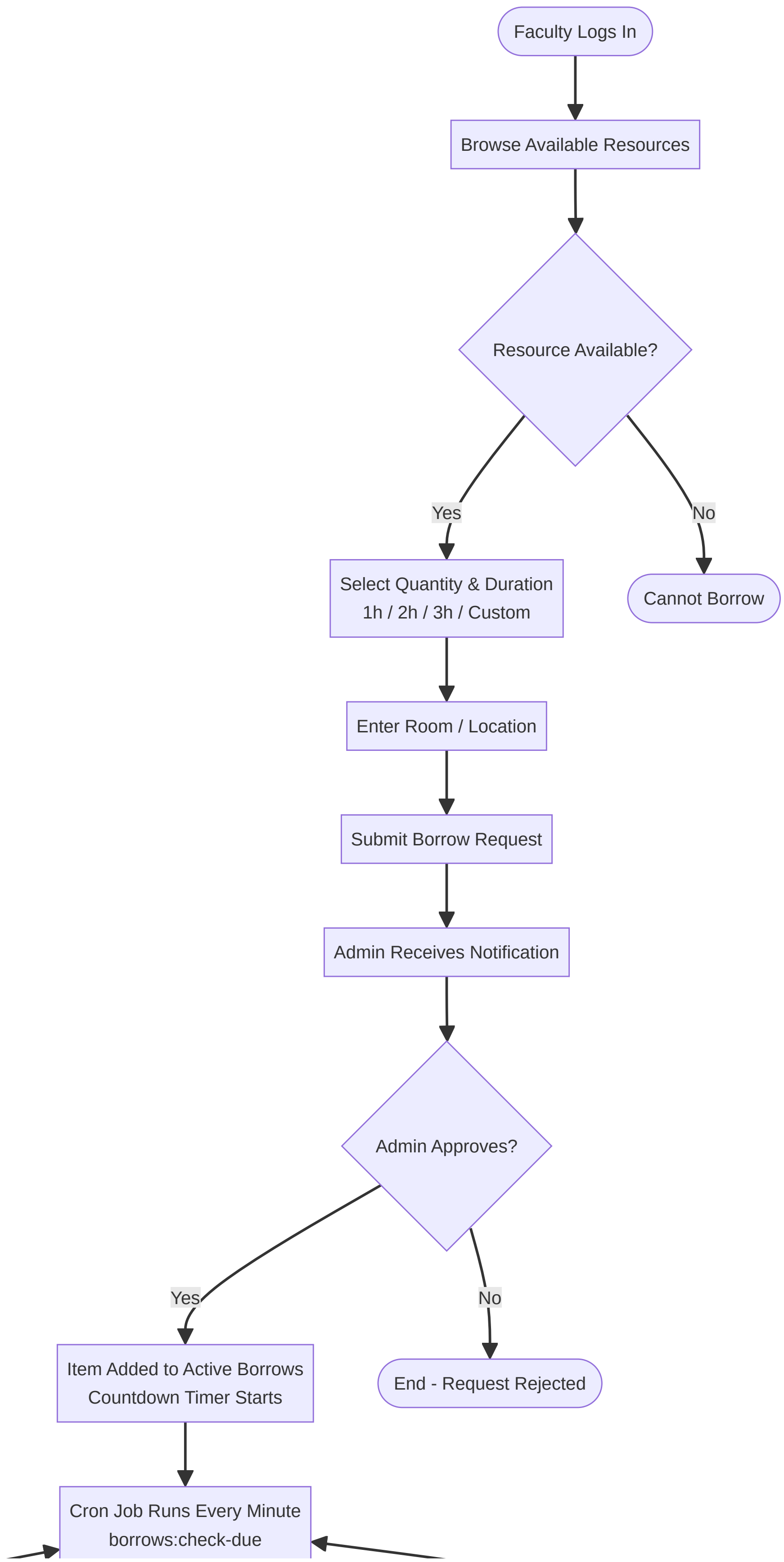
Fig. 2 — Entity Relationship Diagram (ERD) of the Database

E. Technical Stack

- **Backend Framework:** Laravel 11 (PHP 8.2)
- **Database:** MySQL (database name: icct_mis_db)

- **Frontend:** Tailwind CSS utility framework with Alpine.js for interactivity
- **Build Tool:** Vite for asset bundling and hot module replacement
- **Web Server:** Apache 2.4 with HTTPS enforced via Let's Encrypt SSL
- **Transactional Email:** Brevo (formerly SendinBlue) SMTP/API integration
- **API Authentication:** Laravel Sanctum token-based API authentication
- **Task Scheduling:** Laravel Scheduler (cron) running every minute
- **Server:** Ubuntu 22.04 LTS on DigitalOcean (2 vCPU / 4GB RAM)
- **Domain:** https://icct-mis.classapparelph.com

F. System Flowchart


Faculty Logs In
Browse Available Resources
Resource Available?
Yes
No
Select Quantity & Duration
1h / 2h / 3h / Custom
Cannot Borrow
Enter Room / Location
Submit Borrow Request
Admin Receives Notification
Admin Approves?
Yes
No
Item Added to Active Borrows
Countdown Timer Starts
End - Request Rejected
Cron Job Runs Every Minute
borrows:check-due

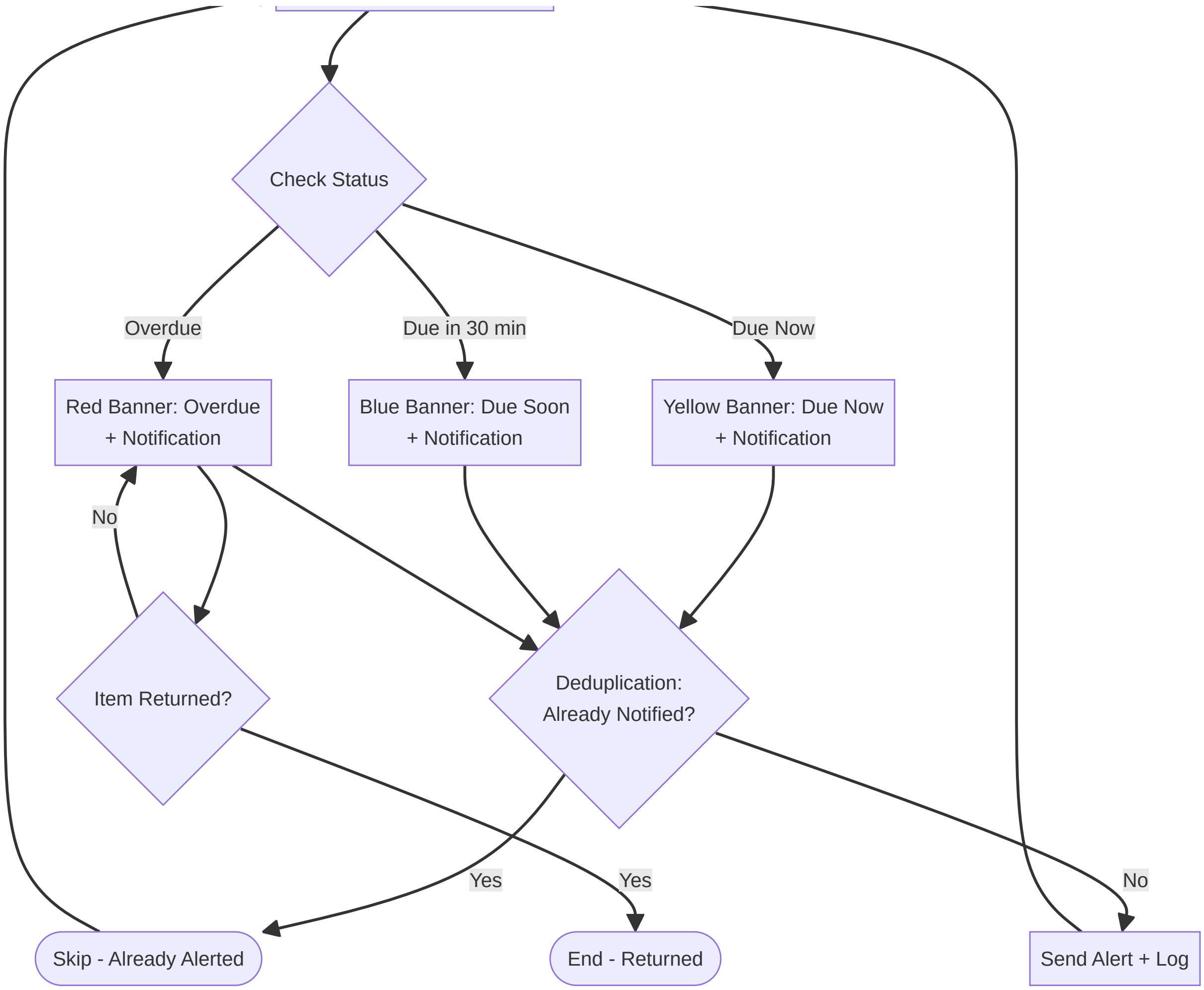


Fig. 3 — System Flowchart: Borrowing Process with Automated Cron Monitoring

G. System Features Implemented

1. Authentication System

Faculty registration with email and password, role-based login (Admin and Faculty), password show/hide toggle, password reset via Brevo transactional email, and email verification support. The system uses Laravel's built-in authentication scaffolding with customized views.

2. Role-Based Access Control

Admin routes protected by AdminMiddleware. Faculty routes are scoped to display only the authenticated user's own data. Different dashboards are rendered for each role, with administrators having access to management panels and analytics.

3. Resource Management (Admin Only)

Administrators can perform full CRUD operations on resources: name, description, total quantity, and availability status. The system auto-calculates available quantity (total quantity minus currently borrowed quantity). Desktop table view adapts to mobile card view for small screens.

4. Borrowing Workflow

Faculty logs in, browses available resources, selects quantity, enters room/location, and chooses a borrowing duration (1 hour, 2 hours, 3 hours, or custom minutes up to 480). The request is submitted as "Pending" status. Administrators review pending requests and can approve (with optional admin notes) or reject (with required reason). Approved items appear on the faculty dashboard with a real-time countdown timer showing remaining time. When the item is returned, the administrator marks it as returned, which records a timestamp and updates resource availability.

5. Dashboard Views

Faculty Dashboard: Welcome message with user name and role, statistics cards (total borrows, active borrows, overdue count), and prominent overdue alert banner when applicable.

Admin Dashboard: Analytics overview showing total resources, available resources, active borrows, pending requests, and overdue items. Weekly transaction chart, most borrowed resources list, and quick action buttons for pending requests.

6. In-App Notifications

Notifications are automatically created for: borrowing request submitted (admin), request approved (faculty), request rejected (faculty), item due now (faculty), item due within 30 minutes (faculty), and item overdue (faculty and admin). Red unread badge appears on the bell icon. Users can mark all as read or delete individual notifications.

7. Automated Due/Overdue Monitoring (Cron)

A custom Artisan command (borrows:check-due) runs every minute via the Laravel Scheduler. It performs three types of checks with deduplication:

- **Due Now:** Items whose due_at time has passed by less than 1 minute since the last notification
- **Due Soon:** Items due within the next 30 minutes (notified once at the 30-minute mark)
- **Overdue:** Items whose due_at is more than 1 minute past, checked every 30 minutes for repeat alerts

Deduplication prevents duplicate notifications by checking the last notification timestamp before sending new alerts.

8. Borrow History

Complete transaction history with summary statistics (total, pending, approved, borrowed, returned, rejected counts). Filterable tabs for all statuses. Desktop view displays a paginated table with color-coded rows (green=returned, yellow=approved/pending, red=overdue) and action buttons for administrators. Mobile view shows card layout with horizontal scroll for longer content.

9. CSV Export (Admin Only)

Administrators can download all borrow records as a CSV file containing: User, Resource, Quantity, Status, Requested Date, Borrowed Date, Due Date, and Returned Date.

10. REST API for Mobile Integration

Complete RESTful API secured with Laravel Sanctum token authentication, ready for Flutter mobile application development. Endpoints include:

- **Authentication:** POST /api/login, POST /api/register, POST /api/logout
- **Dashboard:** GET /api/dashboard
- **Resources:** GET /api/resources
- **Borrows:** GET /api/borrows, POST /api/borrows, PUT /api/borrows/{id}/status
- **Notifications:** GET /api/notifications, PUT /api/notifications/{id}/read
- **Profile:** GET /api/profile, PUT /api/profile
- **User Management (Admin):** GET /api/users, PUT /api/users/{id}/role

11. Settings System

A key-value database store for runtime configuration. Current settings include auto_dismiss_seconds (controls notification auto-dismiss timing). New settings can be added without code changes.

12. Mobile Responsive Design

The interface adapts to three breakpoints: phone (<768px) with card layouts and bottom navigation, tablet (768px–1024px) with compact tables, and laptop (>1024px) with full tables and sidebar navigation. The sidebar features a hamburger toggle menu that opens as an overlay on small screens.

CHAPTER IV: RESULTS AND DISCUSSION

The system was successfully developed, tested, and deployed. All core features described in the objectives and scope were implemented and verified on the live system at https://icct-mis.classapparelph.com. Testing was conducted to validate each functional component. The following table summarizes the testing results:

#	Feature	Status	Result Details
1	Authentication (Register/Login/Password Reset)	✅ Pass	Email/password registration, role-based login, password reset via Brevo email all function correctly
2	Resource Management (CRUD)	✅ Pass	Admin can create, read, update, delete resources; available quantity auto-calculated; both desktop table and mobile card views working
3	Borrowing Workflow (with Room & Duration)	✅ Pass	Faculty selects resource, quantity, room/location, duration; admin approves/rejects; item status tracking works end-to-end
4	In-App Notifications	✅ Pass	Notifications created for approval, rejection, due items; unread badge updates; mark as read and delete functions work
5	Automated Due/Overdue Detection (Cron)	✅ Pass	Cron job runs every minute; correctly detects due now, due soon (30 min), and overdue items; deduplication prevents spam
6	REST API (Sanctum)	✅ Pass	All API endpoints respond correctly with token authentication; login, resources, borrows, notifications endpoints verified
7	Settings System	✅ Pass	Key-value configuration store functions; settings persist in database and are read by the application

8	Brevo Email Integration	✅ Pass	Verification emails and password reset links delivered successfully through Brevo SMTP
9	Mobile Responsive Design	✅ Pass	Properly adapts layout for phone (<768px), tablet (768–1024px), and laptop (>1024px) viewports
10	CSV Export (Admin)	✅ Pass	Administrator can download complete borrow transaction records as comma-separated values

Table 1 — Summary of System Testing Results

All ten test cases passed successfully, confirming that the system meets the objectives outlined in Chapter I. The automated features—particularly the cron-based overdue monitoring with deduplication, the REST API endpoint coverage, and the mobile responsive design—represent significant improvements over the original manual process.

Use Case Diagram

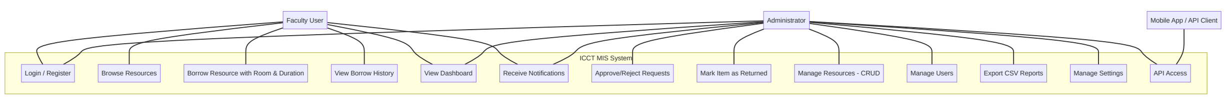


Fig. 4 — Use Case Diagram showing Faculty, Administrator, and Mobile App interactions

CHAPTER V: CONCLUSION AND RECOMMENDATIONS

A. Conclusion

The web-based resource borrowing and returning system was successfully developed and deployed at https://icct-mis.classapparelph.com for the MIS faculty of ICCT Cainta Campus. The system replaces the traditional manual process of submitting request letters or presenting faculty IDs with a streamlined digital workflow. Faculty members can now log in using email and password, browse available resources, submit borrowing requests with duration-based return times and room designation, and receive real-time notifications about their borrowing status. Administrators have access to comprehensive management tools including resource CRUD operations, request approval workflows, borrow history with filterable tabs, CSV reporting, and a dashboard with analytics. The automated cron-based monitoring system ensures that due and overdue items are tracked without manual intervention, with intelligent deduplication to prevent notification fatigue. The REST API provides a foundation for future mobile application development. All ten test cases passed successfully, confirming the system is production-ready and meets the objectives defined in this study.

B. Recommendations

Based on the development and deployment experience, the following enhancements are recommended for future development:

1. **Flutter Mobile Application:** Develop a native Flutter mobile application that consumes the existing REST API, providing push notifications, offline access, and native user experience on smartphones.
2. **QR Code Integration:** Implement the original QR code scanning feature using the existing web-based dashboard as the destination URL, enabling quick resource lookup and streamlined check-in/out through QR codes on resource tags.

3. **Email Notifications:** Extend the existing Brevo integration to send email notifications for borrow-related alerts, ensuring users receive updates even when not logged into the system.
4. **Advanced Analytics:** Add interactive charts, date range filtering, and trend analysis to the admin dashboard for deeper insights into resource utilization.
5. **Resource Categorization:** Implement resource categories and barcode/QR label printing for physical resource tagging.
6. **SMS Notifications:** Add SMS notification support for time-sensitive alerts such as overdue items.
7. **Multi-Campus Scalability:** Extend the system to support multiple campuses and departments beyond the MIS faculty.

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